EVALUATING INTEGRALS

How far up the ladder?

Start with the lightest move and climb only when it does not close the gap.

FIRST Simplify the integrand: expand, split fractions, separate terms.

1 Matches a standard form?
a basic antiderivative applies as it stands

YES → **Antidifferentiate directly**
the fastest path when it works → `Direct antidifferentiation`

NO ↓

2 A standard form in disguise?
complete the square, rewrite constants

YES → **Rewrite, then integrate**
catches near-misses before the heavier tools
→ `Recognizing standard forms`

NO ↓

3 Special structure?
absolute value, piecewise, symmetry

YES → **Use the dedicated shortcut**
split at the sign change or the break; an odd function on [−a, a] gives 0 → `Absolute values`

NO → **Reach for a technique**
substitution, parts, partial fractions, trigonometric substitution → `Integration techniques`

FINALLY Verify: differentiate the antiderivative; for a definite integral, check the sign and the size.